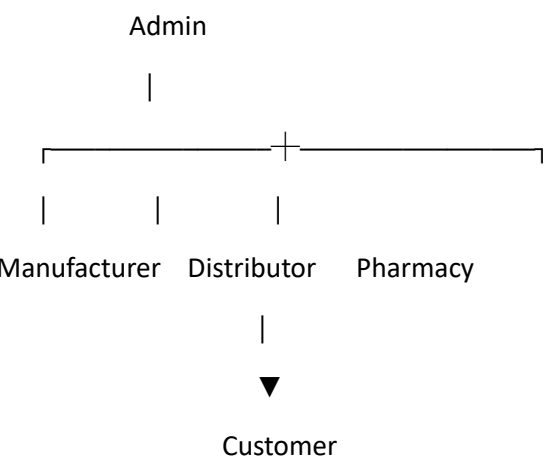


 MediChain v1.0 Architecture

 User Roles



 Medicine Lifecycle

Manufacturer



Create Medicine



Generate Batch Number



Generate QR Code



Blockchain Register



Distributor Receives



Pharmacy Receives

|



Customer Purchases

|



Customer Scans QR

|



Original / Fake Result

Database Design (v1)

Users

Id (GUID)

FullName

Email

PasswordHash

PhoneNumber

Role

IsActive

CreatedAt

UpdatedAt

Medicines

Id (GUID)

MedicineName

BrandName

Description

BatchNumber

ManufactureDate

ExpiryDate

Price

QRCode
BlockchainHash
CurrentOwnerId
Status
CreatedAt
UpdatedAt

OwnershipHistory

Id
MedicineId
FromUserId
ToUserId
TransferDate
TransactionHash

FakeReports

Id
MedicineId
ReportedByUserId
Reason
Status
CreatedAt

AuditLogs

Id
UserId
Action
Description
CreatedAt

 **Relationships**

User

|

└───< Medicines

Medicine

|

└───< OwnershipHistory

User

|

└───< OwnershipHistory

User

|

└───< FakeReports

Medicine

|

└───< FakeReports

 API Structure

/api/auth

POST /register

POST /login

POST /refresh-token

/api/users

GET

GET/{id}

PUT

DELETE

/api/medicines

POST

GET

GET/{id}

PUT

DELETE

/api/ownership

POST /transfer

GET /history/{medicineId}

/api/verification

GET /scan/{qr}

/api/reports

POST

GET

/api/admin

Dashboard

Analytics

🔗 Blockchain (Sprint 6)

Smart Contract Functions:

registerMedicine()

transferOwnership()

verifyMedicine()

reportFakeMedicine()

getMedicineHistory()

 React Pages

Login

Register

Dashboard

Medicine List

Add Medicine

Medicine Details

Transfer Ownership

QR Scanner

Admin Dashboard

Profile

 AI (Final Sprint)

- Fake scan pattern detection
 - Suspicious duplicate QR detection
 - Analytics dashboard
 - Risk score for medicines
-

Sprint Plan

Sprint 1

- API
- Swagger
- GitHub

Sprint 2

- SQL Server
- EF Core
- Migration

Sprint 3 (Next)

- Complete Domain Models
- Entity Relationships
- EF Configurations
- Second Migration

Sprint 4

- Authentication (JWT + BCrypt)

Sprint 5

- Medicine CRUD + QR Generation

Sprint 6

- Blockchain Integration

Sprint 7

- React Frontend

Sprint 8

- AI + Deployment